8			$(x--1)^2+(y-3)^2=r^2$ $r^2=(1--1)^2+(-1-3)^2$ L: $m=-2$ $y=-2x+11$ oe substitution of their $y=-2x+11$ in their $(x+1)^2+(y-3)^2=20$ $x^2-6x+9=0$ oe $(x-3)^2=0$ so repeated root Hence line touches the curve and is a tangent	**M1** **M1** **B1** **B1** **M1** **A1** **E1** **[7]**	**2.1** **1.1** **1.1** **1.1** **1.1** **1.1** **2.4**	Left side correct and = r^2 Or find L first (B1B1), then find equation of line perp to L through (–1, 3) (M1M1) then substitute (M1), solve (A1) then check (E1). soi or $(-6)^2-4\times1\times9=0$	or line through centre which is perpendicular to L has equation $y-3=\frac{1}{2}(x--1)$ meets L at (3, 5) $(3+1)^2+(5-3)^2=r^2=20$ so lines meet at circumference of circle at right angles so L is a tangent